

# Jungseob Lee


NLP Researcher · Ph.D. Student, Korea University

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## Summary


Integrated M.S./Ph.D. researcher at Korea University NLP Lab (Advisor: Prof. Heuiseok Lim), working on large language models and natural language processing — agentic LLMs, LLM efficiency, and efficient reasoning. First-author work at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**; four papers accepted at **EMNLP 2026** (2 Main, 2 Findings). Served as post-training Project Manager (Korea University representative) for **VAETKI**, Korea's sovereign foundation model.


**Research Interests:** Large Language Models, Agentic LLM, LLM Efficiency, Efficient Reasoning

## Education

**Integrated M.S./Ph.D. in Computer Science** *2021 – Present*
*Korea University, Seoul, Korea* **Expected: Feb. 2027**

- NLP Lab, Advisor: Prof. Heuiseok Lim — large language models, agentic LLM, LLM efficiency, and efficient reasoning

## Research Experience

**VAETKI — Korean Sovereign AI Foundation Model** *Jun 2025 – Jan 2026*
*Post-training Project Manager — Korea University Representative* NC AI (NCSOFT)-led consortium

- Led the post-training pipeline (SFT, RL, dataset curation) as PM for Korea University; selected by the Ministry of Science and ICT for Korea's national Sovereign AI Foundation Model program (NC AI-led 14-institution consortium; open-sourced Dec 2025).
- State-of-the-art on Korean language/culture (CLIcK), reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [Model] [Technical Report]

**Cross-lingual Language Transfer for Large Language Models** *2024 – 2025*
*Lead Researcher* (*first author*) Published at ACL 2025 (Oral)

- Cross-lingual optimization that transfers English-centric LLMs to new languages while preserving source-language ability; **Oral** at ACL 2025.
- [PDF]

**Length-aware Byte Pair Encoding for Korean Machine Translation** *2024*
*Lead Researcher* (*first author*) Published at Findings of ACL 2024

- Length-aware BPE tokenization that mitigates over-segmentation in Korean machine translation and improves translation quality; **Findings of ACL 2024**.
- [PDF]

## Awards & Honors

**Outstanding Reviewer Award** *2026*
The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego

**Best Paper Award (Top 1)** *2025*
Annual Conference on Human and Cognitive Language Technology (HCLT 2025)
For "*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*"; **J. Lee** (first author), M. Kim, J. Yoon, *et al.* — Korea University, Soongsil University, Human-Inspired AI Research.

**Minister of Science and ICT Award** *2023*
ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)
For "*KoCheckGPT*", Korean-specialized detection of LLM-generated documents; team M. Kang, **J. Lee**, S. Lee — selected from 79 teams across 28 universities.

## Publications

Underlined = me; * equal contribution. Full list (31 papers) on Google Scholar.

### 2026

- "*The Hallucination Signal Is a Mean Shift: Why Simple Probes Suffice*" — **J. Lee**, J. Seo, H. Lim **EMNLP 2026** (Main) [arXiv]
- "*Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization*" — S. Lee, M. Kim, **J. Lee**, H. Lim **EMNLP 2026** (Main)
- "*DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models*" — **J. Lee**, S. Hong, S. Lee, *et al.* **Findings of EMNLP 2026** [arXiv]
- "*ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models*" — M. Kang*, **J. Lee***, J. Seo, H. Lim **Findings of EMNLP 2026**
- "*Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment*" — S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim **ICLR 2026** (Poster) [PDF]
- "*Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation*" — C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh **Knowledge-Based Systems** [DOI]
- "*Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards*" — **J. Lee**, S. Lee, S. Hong, M. Kim, C. Park, H. Lim Preprint [arXiv]
- "*To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG*" — **J. Lee**, C. Park, H. Lim Preprint [arXiv]
- "*Skin-Deep: A Geometric Diagnostic for Alignment Fragility in LLM Representations*" — D.J. Lee*, **J. Lee***, S. Lee, *et al.* **Findings of AACL-IJCNLP 2026** (*equal contribution) [arXiv]

### 2025

- "*Cross-lingual Optimization for Language Transfer in Large Language Models*" — **J. Lee**, S. Hong, H. Moon, H.S. Lim **ACL 2025** (Oral) [PDF]
- "*An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training*" — S. Son, C. Park, **J. Lee**, *et al.* **Expert Systems with Applications** [DOI]

### 2024

- "*Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation*" — **J. Lee**, H. Moon, S. Lee, C. Park, S. Eo, *et al.* **Findings of ACL 2024** [PDF]
- "*Analysis of Model-, Data-, and User-Centric Approaches for Chat Applications: A Case Study of BlenderBot 2.0*" — C. Park, **J. Lee**, S. Son, *et al.* **Applied Sciences** [DOI]

### 2023

- "*A Survey on Evaluation Metrics for Machine Translation*" — S. Lee, **J. Lee**, H. Moon, *et al.* **Mathematics** (142 citations) [PDF]
- "*PEEP-Talk: A Situational Dialogue-based Chatbot for English Education*" — S. Lee, Y. Jang, C. Park, **J. Lee**, *et al.* **ACL 2023** (Demo) [PDF]

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